

Day 1 Number System - Assignment

- 1) Complete the table with equivalents in Binary, Octal & Hexa for 143.75 (decimal), 11011.101 (binary), 453 (octal), 1F.C (hex)

(i) 143.75 (decimal) to Binary

$$143.75_{(10)} \rightarrow (?)_{(2)}$$

$$\begin{array}{r} 2 \overline{) 143} \\ 2 \overline{) 71} - 1 \\ 2 \overline{) 35} - 1 \\ 2 \overline{) 17} - 1 \\ 2 \overline{) 8} - 1 \\ 2 \overline{) 4} - 0 \\ 2 \overline{) 2} - 0 \\ 1 - 0 \end{array}$$

$$\begin{array}{l} 0.75 \times 2 = 1.5 - 1 \\ 0.5 \times 2 = 1.0 - 1 \\ 0 \times 2 = 0 - 0 \end{array}$$

$$\therefore 143.75_{(10)} = 10001111.110_{(2)}$$

$$143.75_{(10)} \rightarrow (?)_8$$

$$\begin{array}{r} 8 \overline{) 143} \\ 8 \overline{) 17} - 7 \\ 8 \overline{) 2} - 1 \\ 0 - 2 \end{array}$$

$$0.75 \times 8 = 6 - 6$$

$$\therefore 143.75_{(10)} = 217.6_{(8)}$$

$$143.75_{(10)} \rightarrow (?)_{(16)}$$

$$\begin{array}{r} 16 \overline{) 143} \\ 16 \overline{) 8} - 15 (F) \\ 0 - 8 \end{array}$$

$$0.75 \times 16 = 12 - 6$$

$$\therefore 143.75_{(10)} = 8F.C_{(16)}$$

$$(ii) 11011.101_{(2)} \rightarrow (?)_{(10)}$$

$$\Rightarrow 1 \times 2^4 + 1 \times 2^3 + 1 \times 2^2 + 1 \times 2^1 + 1 \times 2^0 + 1 \times 2^{-1} + 1 \times 2^{-2}$$

$$16 + 8 + 4 + 2 + 1 + 0.5 + 0.25 = 27.75$$

$$\therefore 11011.101_{(2)} = 27.75_{(10)}$$

$$\underline{011011.101}_{(2)} \rightarrow (8)$$

$$\therefore 11011.101_{(2)} = 33.5_{(8)}$$

$$\underline{00011011.1010}_{(2)} = 1B.A_{(16)}$$

$$\therefore 11011.101_{(2)} = 1B.A_{(16)}$$

$$(iii) 453_{(8)} \rightarrow (?)_{(10)}$$

$$4 \times 8^2 + 5 \times 8^1 + 3 \times 8^0$$

$$256 + 40 + 3$$

$$453_{(8)} = \underline{\underline{299}}_{(10)}$$

$$\underline{453}_{(8)} \rightarrow (?)_{(2)}$$

$$453_{(8)} = \underline{\underline{100101011}}_{(2)}$$

$$453_{(8)} \rightarrow (?)_{(16)}$$

$$453_{(8)} = \underline{000100101011}_{(2)}$$

$$453_{(8)} = \underline{12B}_{(16)}$$

$$(iv) 1F.C_{(16)} \rightarrow (?)_{(10)}$$

$$1 \times 16^1 + 15 \times 16^0 + 12 \times 16^{-1}$$

$$16 + 15$$

$$1F.C_{(16)} = \underline{31.75}_{(10)}$$

$$1F.C_{(16)} \rightarrow (?)_{(2)}$$

$$1F.C_{(16)} = \underline{00011111.1100}_{(2)}$$

$$1F.C_{(16)} \rightarrow (?)_{(8)}$$

$$1F.C_{(16)} = \underline{00001111.110000}_{(2)}$$

$$\therefore 1F.C_{(16)} = \underline{37.60}_{(8)}$$

Q2 Find equivalents

$$a) 324_{(5)} = (?)_{(10)}$$

$$3 \times 5^2 + 2 \times 5^1 + 4 \times 5^0$$

$$75 + 10 + 4$$

$$\therefore 324_{(5)} = \underline{89}_{(10)}$$

$$b) 32.23_{(4)} = (?)_{(10)}$$

$$3 \times 4^1 + 2 \times 4^0 + 2 \times 4^{-1} + 3 \times 4^{-2}$$

$$= 12 + 2 + 0.5 + 0.1875$$

$$\therefore 32.23_{(4)} = \underline{14.6875}_{(10)}$$

$$c) 231_{(10)} = (?)_{(7)}$$

$$7 \overline{) 231}$$

$$7 \overline{) 33 - 0}$$

$$4 - 5 \uparrow$$

$$\therefore 231_{(10)} = \underline{450}_{(7)}$$

$$d) 189_{(10)} = (?)_{(6)}$$

$$6 \overline{) 189}$$

$$6 \overline{) 31 - 3}$$

$$5 - 1 \uparrow$$

$$\therefore 189_{(10)} = \underline{513}_{(6)}$$

Q3 Find the radix n :

$$a) (312/20)_n = (13.2)_n$$

Taking LHS;

$$(312)_n = 3n^2 + n + 2$$

$$(20)_n = 2n$$

$$\Rightarrow \frac{3n^2 + n + 2}{2n}$$

Taking RHS;

$$(13.2)_n = 1 \times n + 3 + \frac{2}{n}$$

$$= n + 3 + \frac{2}{n}$$

Equating both sides;

$$\frac{3n^2 + n + 2}{2n} = n + 3 + \frac{2}{n}$$

$$3n^2 + n + 2 = 2n^2 + 6n + 2$$

$$n^2 - 5n = 0$$

$$n(n - 5) = 0$$

$$n = 0 \text{ (or) } n = 5$$

$$\therefore n = \underline{\underline{5}}$$

$$b) 264_n + 136_n = 433_n$$

$$\Rightarrow 264_n = 2n^2 + 6n + 4$$

$$136_n = 1n^2 + 3n + 6$$

$$433_n = 4n^2 + 3n + 3$$

$$(2x^2 + 6x + 4) + (x^2 + 3x + 6) = 4x^2 + 3x + 3$$

$$3x^2 + 9x + 10 = 4x^2 + 3x + 3$$

$$x^2 - 6x - 7 = 0$$

$$(x-7)(x+1) = 0$$

$$x=7 \text{ (or)} x=-1$$

$$\underline{x=7} \text{ (}\because \text{radix cannot be negative)}$$

$$c) 221508_{(8)} = 12345_x$$

$$\begin{aligned} \Rightarrow 221508_{(8)} &= 2 \times 8^5 + 2 \times 8^4 + 1 \times 8^3 + 5 \times 8^2 + 5 \times 8^0 \\ &= 2(32768) + 2(4096) + 512 + 5(64) + 5 \\ &= 65536 + 8192 + 512 + 320 + 5 \end{aligned}$$

$$12345_x = 74565_{(10)}$$

$$12345_x = x^4 + 2x^3 + 3x^2 + 4x + 5$$

$$74565 = x^4 + 2x^3 + 3x^2 + 4x + 5$$

$\because$ largest digit is 5; $x > 5$

Let $x=16$;

$$16^4 + 2(16)^3 + 3(16)^2 + 4(16) + 5$$

$$= 65536 + 8192 + 768 + 64 + 5$$

$$= \underline{74565}$$

$$\therefore \underline{x=16}$$

Q4) Addition

a) $1101_{(2)} + 1101111_{(2)}$

$$\Rightarrow \begin{array}{r} 00001101 \\ 11011111 \\ \hline 11101100 \end{array}$$

b) $77652_{(8)} + 76543_{(8)}$

$$\Rightarrow 77652$$

$$76543$$

$$\underline{176415}$$

$$9 = 1 \times 8 + 1$$

Carry Sum

$$12 = 1 \times 8 + 4$$

$$14 = 1 \times 8 + 6$$

$$15 = 1 \times 8 + 7$$

$$\therefore 77652_{(8)} + 76543_{(8)} = \underline{176415}_{(8)}$$

c) $578A_{(16)} + ABDE_{(16)}$

$$\Rightarrow 578A$$

$$ABDE$$

$$\underline{10368}$$

$$A + E = 10 + 14 = 24$$

$$24 = 1 \times 16 + 8$$

$$22 = 1 \times 16 + 6$$

$$19 = 1 \times 16 + 3$$

$$16 = 1 \times 16$$

$$\therefore 578A_{(16)} + ABDE_{(16)} = 10368_{(16)}$$

$$d) 234_{(6)} + 315_{(6)}$$

$$\Rightarrow \begin{array}{r} 234 \\ 315 \\ \hline \end{array}$$

$$553$$

$$\underline{\underline{553}}$$

$$9 = 1 \times 6 + 3$$

$$\therefore 234_{(6)} + 315_{(6)} = 553_{(6)}$$

Q5) Binary $\rightarrow$ Gray code :

$$a) 10101_{(2)}$$

$$\Rightarrow \text{Binary : } 1 \ 0 \ 1 \ 0 \ 1$$

$$\text{Gray : } 1 \ 1 \ 1 \ 1 \ 1$$

$$1 \oplus 0 = 1$$

$$0 \oplus 1 = 1$$

$$1 \oplus 0 = 1$$

$$0 \oplus 1 = 1$$

$$\therefore 10101_{(2)} = 11111_{(Gray)}$$

b) 01110₍₂₎ $\Rightarrow$ Binary : 0 1 1 1 0

Gray : 1 0 0 1 0

$1 \oplus 0 = 1$

$1 \oplus 1 = 0$

$1 \oplus 1 = 0$

$0 \oplus 1 = 1$

$\therefore 01110_{(2)} = 01001_{(Gray)}$

Q6 Gray code $\rightarrow$ Binarya) 01101_(Gray) $\Rightarrow$ Gray : 0 1 1 0 1

Binary : 0 1 0 0 1

 $\rightarrow$

$0 \oplus 1 = 1$

$1 \oplus 1 = 0$

$0 \oplus 0 = 0$

$0 \oplus 1 = 1$

$\therefore 01101_{(Gray)} = 01001_{(2)}$

b) 10101 (Gray)

$\Rightarrow$ Gray : 1 0 1 0 1

Binary : 1 1 0 0 1

$$1 \oplus 0 = 1$$

$$0 = 1 \oplus 1$$

$$1 \oplus 1 = 0$$

$$0 = 1 \oplus 1$$

$$0 \oplus 0 = 0$$

$$1 = 1 \oplus 0$$

$$0 \oplus 1 = 1$$

$$\therefore 10101 \text{ (Gray)} = \underline{\underline{11001}} \text{ (2)}$$